Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A cylinder of capacity $20\ \mathrm{L}$ is filled with H_2 gas. The total average kinetic energy of translatory motion of its molecules is $1.5 \times 10^5\ \mathrm{J}$. The pressure of hydrogen in the cylinder is

(A) $2 \times 10^6\ \mathrm{N/m^2}$

(B) $3 \times 10^6\ \mathrm{N/m^2}$

(C) $4 \times 10^6\ \mathrm{N/m^2}$

(D) $5 \times 10^6\ \mathrm{N/m^2}$

Q.2

An air bubble of volume V_0 is released by a fish at a depth h in a lake. The bubble rises to the surface. Assume constant temperature and standard atmospheric pressure P above the lake. The volume of the bubble just before touching the surface will be (density of water is ρ)

(A) V_0

(B) $V_0(\rho gh/P)$

(C) $\frac{V_0}{\left(1+\frac{\rho gh}{P}\right)}$

(D) $V_0\left(1+\frac{\rho gh}{P}\right)$

Q.3

For a gas the difference between the two specific heats is $4150\ \mathrm{J/kgK}$. What is the specific heat at constant volume of gas if the ratio of specific heats is 1.4

(A) $8475\ \mathrm{J/kg-K}$

(B) $5186\ \mathrm{J/kg-K}$

(C) $1660\ \mathrm{J/kg-K}$

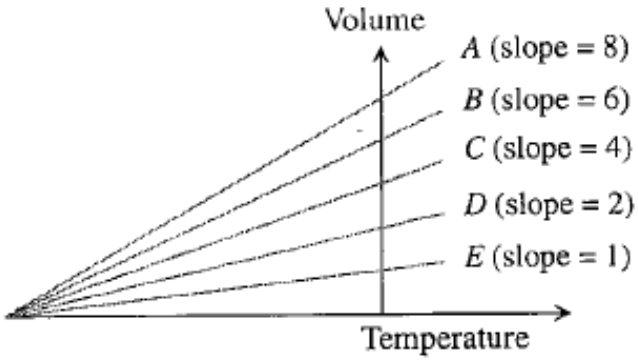
(D) $10375\ \mathrm{J/kg-K}$

Q.4

The expansion of an ideal gas of mass m at a constant pressure P is given by the straight line D. Then the

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expansion of the same ideal gas of mass $2m$ at a pressure $P/2$ is given by the straight line



(A) E

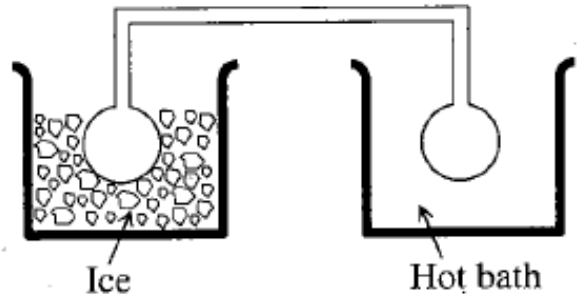
(B) C,

(C) B

(D) A

Q.5

Two identical glass bulbs are interconnected by a thin glass tube. A gas is filled in these bulbs at NTP. If one bulb is placed in ice and another bulb is placed inside hot bath, then the pressure of the gas becomes 1.5 times. The temperature of hot bath will be



(A) $100^{\circ}\mathrm{C}$

(B) $182^{\circ}\mathrm{C}$

(C) $256^{\circ}\mathrm{C}$

(D) $546^{\circ}\mathrm{C}$

Q.6

The temperature of a gas is raised while its volume remains constant, the pressure exerted by the gas on the walls of the container increases because its molecules

(A) lose more kinetic energy to the wall

(B) are in contact with the wall for a shorter time

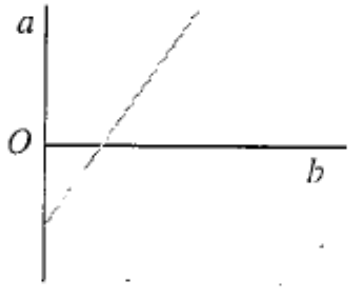
(C) strike the wall more often with higher velocities

(D) collide with each other with less frequency

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Q.7

The expansion of unit mass of a perfect gas at constant pressure is shown in Fig. Here



(A) $a =$ volume, $b = {}^{\circ}\mathrm{C}$ temperature

(B) $a =$ volume, $b = \mathrm{K}$ temperature

(C) $a = {}^{\circ}\mathrm{C}$ temperature, $b =$ volume

(D) $a = \mathrm{K}$ temperature, $b =$ volume

Q.8

A gas is heated at a constant pressure. The fraction of heat supplied used for external work is

(A) $\frac{1}{\gamma}$

(B) $\left(1-\frac{1}{\gamma}\right)$

(C) $\gamma - 1$

(D) $\left(1-\frac{1}{\gamma^2}\right)$

Q.9

Energy of all molecules of a monatomic gas having a volume V and pressure P is $3/2PV$. The total translational kinetic energy of all molecules of a diatomic gas at the same volume and pressure is

(A) $1/2PV$

(B) $3/2PV$

(C) $5/2PV$

(D) $3PV$

Q.10

The average degrees of freedom per molecule for a gas are 6 . The gas performs $25\ \mathrm{J}$ of work when it expands at constant pressure. The heat absorbed by gas is

(A) $75\ \mathrm{J}$

(B) $100\ \mathrm{J}$

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(C) $150\ \text{J}$

(D) $125\ \text{J}$

Q.11

A box contains N molecules of a perfect gas at temperature T_1 and pressure P_1. The number of molecules in the box is doubled keeping the total kinetic energy of the gas same as before. If the new pressure is P_2 and temperature T_2, then

(A) $P_2 = P_1, T_2 = T_1$

(B) $P_2 = P_1, T_2 = \frac{T_1}{2}$

(C) $P_2 = 2P_1, T_2 = T_1$

(D) $P_2 = 2P_1, T_2 = \frac{T_1}{2}$

Q.12

A gas is filled in the cylinder shown in the figure. The two pistons are joined by a string. If the gas is heated, the pistons will



(A) move towards left

(B) move towards right

(C) remain stationary

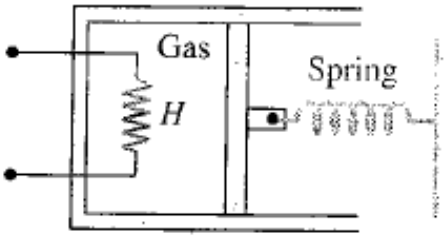
(D) none of these

Q.13

An ideal monatomic gas is confined in a cylinder by a spring-loaded piston of cross-section $8 \times 10^{-3}\ \text{m}^2$. Initially the gas is at $300\ \text{K}$ and occupies a volume of $2.4 \times 10^{-3}\ \text{m}^3$ and the spring is in a relaxed state. The gas is heated by a small heater coil H. The force constant of the spring is $8000\ \text{N/m}$ and the atmospheric pressure is $1.0 \times 10^5\ \text{Pa}$. The cylinder and piston are thermally insulated The piston and the spring are massless and there is no friction between the piston and cylinder. There is no heat loss through heater coil wire leads and thermal capacity of the heater coil is negligible. With all the above assumptions, if the gas is

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heated by the heater until the piston moves out slowly by $0.1\ \mathrm{m}$, then the final temperature is



(A) $400\ \mathrm{K}$

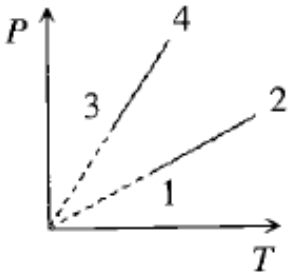
(B) $800\ \mathrm{K}$

(C) $1200\ \mathrm{K}$

(D) $300\ \mathrm{K}$

Q.14

Pressure versus temperature graph of an ideal gas of equal number of moles of different volumes is plotted as shown in Fig. Choose the correct alternative.



(A) $V_1 = V_2, V_3 = V_4$ and $V_2 > V_3$

(B) $V_1 = V_2, V_3 = V_4$ and $V_2 < V_3$

(C) $V_1 = V_2 = V_3 = V_4$

(D) $V_4 > V_3 > V_2 > V_1$

Q.15

If pressure of a gas contained in a closed vessel is increased by 0.4% when heated by $1^\circ\mathrm{C}$, the initial temperature must be

(A) $250\ \mathrm{K}$

(B) $250^\circ\mathrm{C}$

(C) $2500\ \mathrm{K}$

(D) $25^\circ\mathrm{C}$

Q.16

The capacity of a vessel is $3\ \mathrm{L}$. It contains $6\ \mathrm{g}$ oxygen, $8\ \mathrm{g}$ nitrogen and $5\ \mathrm{g}\ \mathrm{CO_2}$ mixture at $27^\circ\mathrm{C}$. If $R = 8.31\ \mathrm{J/molK}$, then the pressure in the vessel in $\mathrm{N/m^2}$ will be (approx.)

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(A) 5×10^5

(B) 5×10^4

(C) 10^6

(D) 10^5

Q.17

The specific heat at constant volume for the monatomic argon is $0.075 \mathrm{kcal/kg-K}$, whereas its gram molecular specific heat is $C_v = 2.98 \mathrm{cal/mol/K}$. The mass of the argon atom is (Avogadro's number $= 6.02 \times 10^{23}$ molecules $/\mathrm{mol}$)

(A) 6.60×10^{-23} g

(B) 3.30×10^{-23} g

(C) 2.20×10^{-23} g

(D) 13.20×10^{-23} g

Q.18

Forty calories of heat is needed to raise the temperature of $1 \mathrm{~mol}$ of an ideal monatomic gas from $20^{\circ} \mathrm{C}$ to $30^{\circ} \mathrm{C}$ at a constant pressure. The amount of heat required to raise its temperature over the same interval at a constant volume $\left(R = 2 \mathrm{calmol}^{-1} \mathrm{~K}^{-1}\right)$ is

(A) $20 \mathrm{cal}$

(B) $40 \mathrm{cal}$

(C) $60 \mathrm{cal}$

(D) $80 \mathrm{cal}$

Q.19

Hydrogen gas is filled in a balloon at $20^{\circ} \mathrm{C}$. If temperature is made $40^{\circ} \mathrm{C}$, pressure remaining same, what fraction of hydrogen will come out?

(A) 0.07

(B) 0.25

(C) 0.5

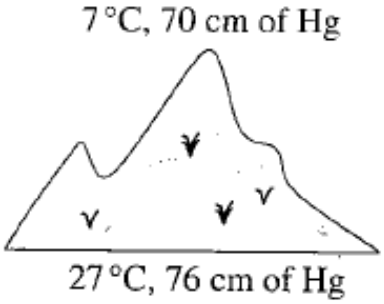
(D) 0.75

Q.20

At the top of a mountain a thermometer reads $7^{\circ} \mathrm{C}$ and a barometer reads $70 \mathrm{~cm}$ of Hg. At the bottom of the mountain these read $27^{\circ} \mathrm{C}$ and $76 \mathrm{~cm}$ of Hg, respectively. Ratio of density of air at the top with that of

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bottom is



(A) $75/76$

(B) $70/76$

(C) $76/75$

(D) $76/70$

Q.21

A long container has air enclosed inside at room temperature and atmospheric pressure $\left(10^5\ \mathrm{Pa}\right)$. It has a volume of $20000\mathrm{cc}$. The area of cross section is $100\ \mathrm{cm}^2$ and force constant of spring is $k_{\text{spring}} = 1000\ \mathrm{N/m}$. We push the right piston isothermally and slowly till it reaches the original position of the left piston which is movable. Final length of air column is found to be $25h\ \mathrm{cm}$. Assume that spring is initially relaxed. Find h.

Q.22

If $P - V$ diagram of a diatomic gas is plotted, it is a straight line passing through origin. The molar heat capacity of the gas in the process is nR where n is an integer. Find value of n.

Q.23

A vessel of volume $0.2\ \mathrm{m}^3$ contains hydrogen gas at temeperature $300\ \mathrm{K}$ and pressure $1\mathrm{bar}$. Find the heat (in Kcal) required to raise the temperature to $400K$. The molar heat capacity of hydrogen at constant volume is $5\mathrm{cal/mol}K$.

Q.24

A vessel of volume $3V$ contains a gas at pressure $4P_0$ and another vessel of volume $2V$ contains the same gas at pressure $1.5P_0$. Both vessels have same temperature. When both vessels are connected by a tube of negligible volume, the equilibrium is IP_0, where I is an integer. Find the value of I.

Q.25

A vessel contains 1 mole O_2 gas at a temperature T. An identical vessel contains one mole of He gas at a temperature $2T$. Find the ratio of their pressure (He to O_2).

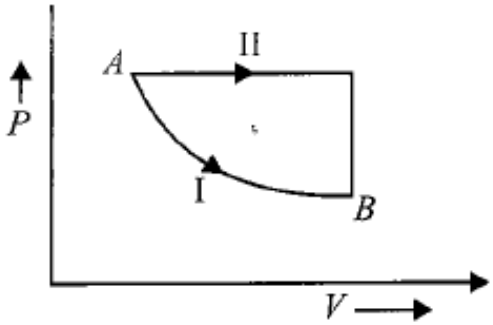
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Q.26

Calculate the pressure exerted by a mixture of $8\ \mathrm{g}$ of oxygen, $14\ \mathrm{g}$ of nitrogen and $22\ \mathrm{g}$ of carbon dioxide in a container of $10\ \mathrm{L}$ at a temperature of $27^{\circ}\mathrm{C}$ [in $10^5\ \mathrm{N/m^2}$ and to the nearest integer].

Q.27

A certain mass of gas is taken from an initial thermodynamics state A to another state B by process I and II. In process I for the gas does $5\ \mathrm{J}$ of work and absorbs $4\ \mathrm{J}$ of heat energy. In process II, the gas absorbs 5 joules of heat. The work done by the gas in process II is



Q.28

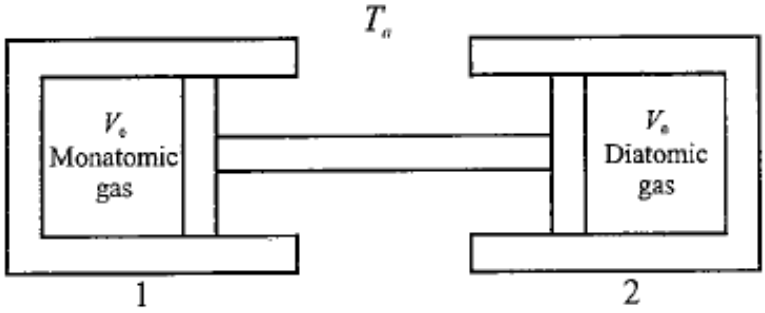
A vessel contains helium, which expands at constant pressure when $15\ \mathrm{kJ}$ of heat is supplied to it. What will be the variation of the internal energy of the gas? (in kJ)

Q.29

A vessel contains a mixture of one mole of oxygen and two moles of nitrogen at $300\ \mathrm{K}$. Find the ratio of average rotational kinetic energies per $\mathrm{O_2}$ molecule to per $\mathrm{N_2}$ molecule.

Q.30

The two conducting cylinder-piston systems shows below are linked. Cylinder 1 is filled with a certain molar quantity of a monatomic ideal gas, and cylinder 2 is filled with an equal molar quantity of a diatomic ideal gas. The entire apparatus is situated inside an oven whose temperature is $T_a = 27^{\circ}\mathrm{C}$. The cylinder volumes have the same initial value $V_0 = 100\mathrm{cc}$. When the oven temperature is slowly raised to $T_b = 127^{\circ}\mathrm{C}$. What is the volume change ΔV (in cc) of cylinder 1?



Answers & Solutions

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Q.1 Answer:

$5 \times 10^6 \text{ N/m}^2$

Solution:

Kinetic energy $E = 1.5 \times 10^5$ **J, volume,** $V = 20L = 20 \times 10^{-3} \text{ m}^3$

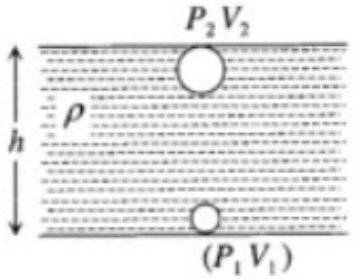
Pressure

$= \frac{2}{3}\frac{E}{V} = \frac{2}{3}\left(\frac{1.5\times10^5}{20\times10^{-3}}\right) = 5 \times 10^6 \text{ N/m}^2$

Q.2 Answer:

$V_0\left(1 + \frac{\rho gh}{P}\right)$

Solution:



If P **is the atmospheric pressure at the top of the lake and the volume of bubble just before touching surface is** V**, then from** $P_1V_1 = P_2V_2$

$$(P + h\rho g)V_0 = PV \quad \Rightarrow V = \left(\frac{P + h\rho g}{P}\right)V_0$$

$$\therefore V = V_0\left[1 + \frac{\rho gh}{P}\right]$$

Q.3 Answer:

$10375 \text{ J/kg} - \text{K}$

Solution:

Given $\quad c_p - c_r = 4150$

and $c_\rho / c_v = 1.4 \Rightarrow c_p = 1.4c_r$

By substituting the value of c_p **in Eq. (i), we get**

$$1.4c_r - c_r = 4150$$
$$\Rightarrow 0.4c_r = 4150$$

Q.4 Answer:

A

Solution:

For the first condition slope (M/P) **graph is** D **(given in the problem)**

For the second condition slope

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$\frac{2M}{P/2} = 4\left(\frac{M}{P}\right)$

i.e., slope becomes four times so graph A is correct in this condition.

Q.5 Answer:

546°C

Solution:

Quantity of gas in these bulbs is constant, i.e., initial no, of moles in both the bulbs = final number of moles

$\mu_1 + \mu_2 = \mu_1' + \mu_2'$

$\frac{PV}{R(273)} + \frac{PV}{R(273)} = \frac{1.5PV}{R(273)} + \frac{1.5PV}{R(T)}$

$\Rightarrow \frac{2}{273} = \frac{1.5}{273} + \frac{1.5}{T}$

$\Rightarrow T = 819\ \text{K} = 546^\circ\text{C}$

Q.6 Answer:

strike the wall more often with higher velocities

Solution:

Due to increase in temperature root-mean-square velocity of gas molecules increases. So they strike the wall more often with higher velocity. Hence the pressure exerted by a gas on the walls of the container increases.

Q.7 Answer:

$a = {}^\circ\text{C}$ temperature, $b =$ volume

Solution:

In the given graph, line has a positive slope with the x-axis and negative intercept on the y-axis.

So we can write the equation of line as

$y = mx - c$ **...(i)**

According to Charles's law

$V_t = \frac{V_0}{273}t + V_0$

by rewriting this equation, we get

$t = \left(\frac{273}{V_0}\right)V_t - 273$ **...(ii)**

Q.8 Answer:

$\left(1 - \frac{1}{\gamma}\right)$

Solution:

We know fraction of given energy that goes to increase the internal energy $= 1/\gamma$

So we can say the fraction of given energy that is supplied for external work $= 1 - (1/\gamma)$.

Q.9 Answer:

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$3/2PV$

Solution:

Energy of 1 mol of gas

$= \frac{f}{2}RT = \frac{f}{2}PV$

where f = degree of freedom

Monatomic or diatomic gases possess equal degree of freedom for translational motion and that is equal to 3 , i.e., $f = 3$

$E = \frac{3}{2}PV$

Q.10 Answer:

100 J

Solution:

As $f = 6$ (given), therefore

$\gamma = 1 + \frac{2}{f} = 1 + \frac{2}{6} = \frac{4}{3}$

Fraction of energy given for extemal work

$\frac{\Delta W}{\Delta Q} = \left(1 - \frac{1}{\gamma}\right)$

Q.11 Answer:

$P_2 = P_1, T_2 = \frac{T_1}{2}$

Solution:

Kinetic energy of N molecules of gas,

$E = \frac{3}{2}NkT$

Initially

$E_1 = \frac{3}{2}N_1kT_1$

and finally

$E_2 = \frac{3}{2}N_2kT_2$

But according to the problem $E_1 = E_2$ and $N_2 = 2N_1$

$\frac{3}{2}N_1kT_1 = \frac{3}{2}(2N_1)kT_2$

$\Rightarrow T_2 = \frac{T_1}{2}$

Since the kinetic energy is constant

$$\frac{3}{2}N_1kT_1 = \frac{3}{2}N_2kT_2$$

$\Rightarrow N_1T_1 = N_2T_2$

$\therefore \quad NT = \text{ constant}$

From ideal gas equation of N molecules $PV = NkT$

$\Rightarrow P_1V_1 = P_2V_2$

$P_1 = P_2$

$(\text{as } V_1 = V_2 \quad \text{and} \quad NT = \text{constant})$

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Q.12 Answer:

move towards right

Solution:

When the gas is heated, its temperature increases and thus it expands. As the cross-sectional area of the right piston is more, therefore greater force will work on it (as Force = Pressure x Area). So the piston will move towards the right.

Option B is correct.

Q.13 Answer:

800 K

Solution:

$$V_1 = 2.4 \times 10^{-3}\ \mathrm{m}^3, \quad P_1 = P_0 = 10^5\ \mathrm{N/m}^2$$
$$\text{and} \quad T_1 = 300\ \mathrm{K} \quad \text{(given)}$$

If area of cross section of piston is A and it moves through distance x, then increment in volume of the gas $= Ax$.

If force constant of a spring is k, then force $F = kx$ and pressure $= F/A = kx/A$

$$V_2 = V_1 + Ax = 2.4 \times 10^{-3} + 8 \times 10^{-3} \times 0.1 = 3.2 \times 10^{-3}$$

Q.14 Answer:

$V_1 = V_2, V_3 = V_4$ and $V_2 > V_3$

Solution:

From ideal gas equation,

$$PV = \mu RT$$
$$P = \frac{\mu R}{V} T$$

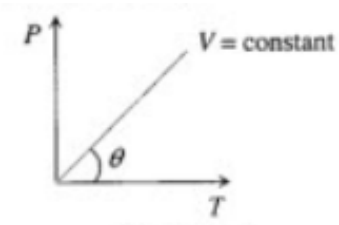
Comparing, this equation with $y = mx$

Slope of line, $\tan\theta = m = \mu R/V$

i.e. $V \propto \frac{1}{\tan\theta}$

It means line of smaller slope represents greater volume of gas.

For the given problem figure



Points 1 and 2 are on the same line, so they will represent same volume, i.e., $V_1 = V_2$.

Similarly points 3 and 4 are on the same line, so they will represent same volume, i.e., $V_3 = V_4$.

But $V_1 > V_3\ (= V_4)$ or $V_2 > V_3\ (= V_4)$ as slope of line

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Q.15 Answer:

250 K

Solution:

$P_1 = P, \quad T_1 = T, \quad P_2 = P + (0.4\% \text{ of } P)$
$\Rightarrow P_2 = P + \frac{0.4}{100}P = P + \frac{P}{250}$ and $T_2 = T + 1$

From Gay-Lussac's law

$\frac{P_1}{P_2} = \frac{T_1}{T_2}$
$\Rightarrow \frac{P}{P+\frac{P}{250}} = \frac{T}{T+1}$
(as $V =$ constant for closed vessel)

By solving we eet $T = 250$ K

Q.16 Answer:

5×10^5

Solution:

By Dalton's law

$$P = P_1 + P_2 + P_3 = \frac{\mu_1 RT}{V} + \frac{\mu_2 RT}{V} + \frac{\mu_3 RT}{V}$$
$$= \frac{RT}{V}[\mu_1 + \mu_2 + \mu_3] = \frac{RT}{V}\left[\frac{m_1}{M_1} + \frac{m_2}{M_2} + \frac{m_3}{M_3}\right]$$

Q.17 Answer:

6.60×10^{-23} g

Solution:

Molar specific heat = molecular weight × gram specific heat

$$C_v = M \times c_v$$
$$2.98\text{cal/mol} - \text{K} = M \times 0.075\text{kcal/kg} \cdot \text{K}$$
$$= M \times \frac{0.075 \times 10^3}{10^3}\text{cal/g} - \text{K}$$

∴ Molecular weight of argon

$M = \frac{2.98}{0.075} = 39.7$ g

i.e., mass of 6.023×10^{23} **atom** $= 39.7$ g,

Therefore, mass of single atom

$= \frac{39.7}{6.023 \times 10^{23}} = 6.60 \times 10^{-23}$ g

Q.18 Answer:

20cal

Solution:

At constant pressure, the amount of heat required is

$\mathrm{Q = nC_p\Delta T}$

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$40 = 1 \times C_p \times 10$

Thus, $C_p = 4$

$\therefore C_v = C_p - R$

$= 4 - 2 = 2\text{cal/mol} - \text{K}$

Now $(\Delta Q)_r = \mu C_v \Delta T$

$= 1 \times 2 \times (30 - 20) = 20\text{cal}$

Q.19 Answer:

0.07

Solution:

As $V \propto T$

$\therefore \frac{V_2}{V_1} = \frac{T_2}{T_1}$

$\Rightarrow V_2 = \left(\frac{313}{293}\right) V_1$

Fraction of gas that comes out

$= \frac{V_2 - V_1}{V_1} = \frac{\left(\frac{313}{293}\right)V_1 - V_1}{V_1} = \frac{20}{293} = 0.07$

Q.20 Answer:

75/76

Solution:

Ideal gas equation, in terms of density,

$\frac{P_1}{\rho_1 T_1} = \frac{P_2}{\rho_2 T_2} =$ **constant**

$\therefore \quad \frac{\rho_1}{\rho_2} = \frac{P_1}{P_2} \times \frac{T_2}{T_1}$

Q.21 Answer:

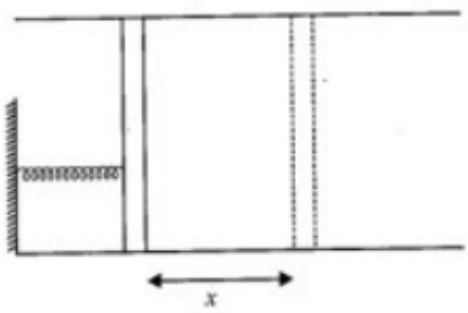
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Solution:

$P_2 V_2 = P_1 V_1$

$V_1 = 20000\text{cc} \quad P_1 = 10^5\ \text{Pa}$

$V_2 = Ax, \quad P_2 = P_0 + \frac{kx}{A}$



$\Rightarrow 10^5 \times 20000 \times 10^{-6}$

$= \left(10^5 + \frac{1000x}{100 \times 10^{-4}}\right) 100x \times 10^{-4}$

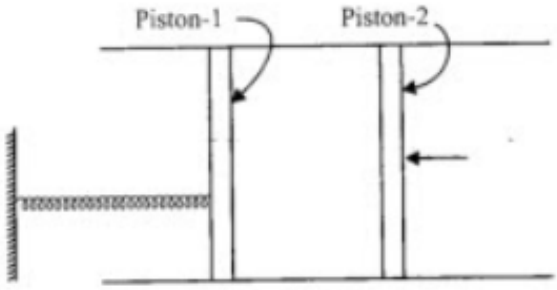
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$\Rightarrow \quad 2000 = 1000[x(1+x)]$

$x^2 + x - 2 = 0$

$x = \frac{-1 \pm \sqrt{1+8}}{2} = 1 \text{ m} = 100 \text{ cm}$

$\Rightarrow 25h = 100 \Rightarrow h = 4$



Q.22 Answer:

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Solution:

$P \propto V$ **since graph is a straight line** $PV^{-1} =$ **constant**

Q.23 Answer:

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Solution:

As here volume of gas remains constant, $(\Delta Q)_V = \mu C_1 \Delta T,$

Here $C_V = 5\text{cal/molK}$

And $\Delta T = (400 - 300) = 100 \text{ K}$

And so for ideal gas $PV = \mu RT,$

$\mu = \frac{(10)^5 \times (0.2)}{8.31 \times 300} = 8 \text{ mol}$

$(\Delta Q)_v = 8 \times 5 \times 100 = 4\text{kcal}$

Q.24 Answer:

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Solution:

No change of temperature-

$\text{So, } (n_1 + n_2)_i = (n_1 + n_2)_f$

$\frac{PV_1}{RT} + \frac{P_2V_2}{RT} = \frac{PV_1}{RT} + \frac{PV_2}{RT}$

$\text{So, } P = \frac{P_1V_1 + P_2V_2}{V_1 + V_2}$

$= \frac{(4P_0)(3V) + (1.5P_0)(2V)}{5V} = 3P_0 = IP_0$

So, $l = 3$

Q.25 Answer:

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Solution:

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Using $P = \frac{nRT}{V}$, we get
$\frac{P_{He}}{P_{O_2}} = \frac{2T}{V}\frac{V}{T} = 2$

Q.26 Answer:

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Solution:

Since $PV = nRT = \frac{m}{M}RT$ we have
$P = \frac{RT}{V}\left[\frac{m_1}{M_1} + \frac{m_2}{M_2} + \frac{m_3}{M_3}\right]$
where m 's are mass of given gas and M is its molecular mass
$P = \frac{8.3\times300}{10\times10^{-3}}\left[\frac{8}{32} + \frac{14}{28} + \frac{22}{44}\right]$
$= 3.11 \times 10^5 \text{ N/m}^2$
In 10^5 N/m^2, $P = 3$

Q.27 Answer:

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Solution:

For process I : $dQ = dU + dW$
$\Rightarrow 4 = dU + 5 \Rightarrow dU = -1$
For process II : $5 = dU + dW$
$\Rightarrow 5 = -1 + dW \Rightarrow dW = +6$ J

Q.28 Answer:

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Solution:

Here $\Delta Q = 15000$ J (given)
In an isobaric process?
$\Delta Q = nC_p\Delta T$, $\Delta U = nC_1\Delta T$ (always)
$\frac{\Delta U}{\Delta Q} = \frac{nC_V\Delta T}{nC_p\Delta T} = \frac{1}{\gamma} \Rightarrow \Delta U = \frac{\Delta Q}{\gamma}$
or $\Delta U = \frac{3}{5} \times 15 = 9$ kJ

Q.29 Answer:

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Solution:

Both gases will have same number of rotational degree of freedom (equal to 2). Since both are diatomic and are at the same temperature. Average rotational kinetic energy per molecule for both the gases $= 2\left(\frac{1}{2}KT\right)$. Required ratio = 1

Q.30 Answer:

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Solution:

$(0)\ P_0V_0 = nRT_A$

$(P_0 + \Delta P)(V_0 + \Delta V) = nRT_b$

$(P_0 + \Delta P)(V_0 - \Delta V) = nRT_b$

$\Delta V = 0$

Pressure depends on number of moles and is independent of nature of gas.

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